Intermediate Macroeconomics

BSc. in Business Administration and BSc. in Finance and Accounting

Intermediate Macroeconomics – BMAK

Chapter 5: The Macroeconomy in the Medium Run

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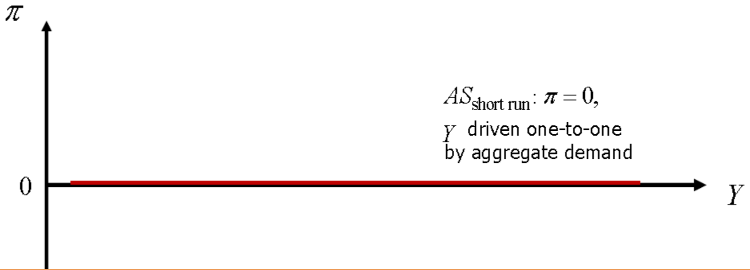
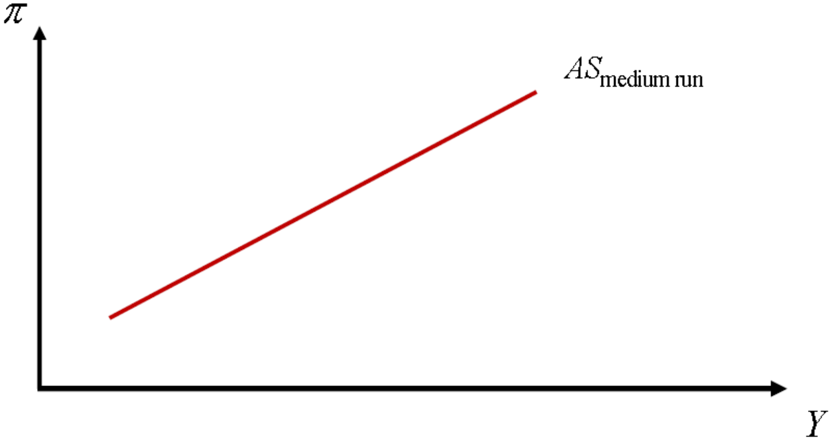
- The model of business cycles we have developed in the previous chapter of the course was a short-run model.
- As we emphasized, in the short-run model, aggregate output supply responds one-to-one to changes in aggregate output demand, with prices being fixed.
- We now turn to studying the macroeconomy in the **medium run**. In the medium-run model, firms will generally respond to changes in aggregate output demand by adjusting both output supply and prices.
- The medium run is implicitly defined as that time period during which macroeconomic outcomes adjust from their short-run to their long-run values. (The long-run time paths are not part of business cycles and will be studied when we turn to the analysis of economic growth.)
- To obtain the medium-run macroeconomic outcomes, we will need to understand how prices change with the level of output produced.
- This **"aggregate supply" relationship** *(AS-curve)* is typically plotted in diagrams with the growth rate of prices (that is, the rate of inflation) on the vertical axis, and the level of output on the horizontal axis.
- In the short-run model, this relationship would be depicted by a horizontal line at zero



+ In the medium-run model, the aggregate supply relationship will, in contrast, turn out to be:

$$\pi = \underbrace{\pi^e}_{\substack{\text{expected rate}\\ \text{of inflation}}} + \underbrace{\omega}_{\omega>0} \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \underbrace{s}_{\substack{\text{shock to the cost}\\ \text{of production}\\ \text{("supply" shock)}}}, \tag{1}$$

yielding the following upward sloping graph of the medium-run *AS*-curve:



To derive this medium-run *AS*-curve, we will need to model the pricing decisions of firms, specifically how these entail responses to changes in their cost of production. **The Battle of Mark-Ups**

+ In the medium run, aggregate price dynamics are in large parts driven by
  - firms with market power (such market power often achieved through product differentiation) attempting to set prices as a mark-up over costs of production (including wages), and
  - employees or their wage negotiators attempting to increase real wages by pushing up nominal wages relative to expected prices and productivity.

Firms' Pricing

+ We model firms as making their production decisions so as to maximize revenue relative to the costs of production:

$$\max_{Y} \underbrace{P}_{\substack{\text{s.t. } P=P(Y) \\ \text{the inverse demand function}}} \cdot Y - \underbrace{W}_{\substack{\text{nominal wages, taken as} \\ \text{given by firms when} \\ \text{deciding about medium-run} \\ \text{output supply and its pricing}}} \cdot \underbrace{L}_{\text{labor hours}} - \underbrace{r^k \cdot K}_{\substack{\text{capital stock and rental rate} \\ \text{thereon, both taken as given} \\ \text{by firms when deciding about} \\ \text{medium-run output supply} \\ \text{and its pricing}}}, \tag{2}$$

which implies the first-order condition

$$\begin{aligned} & \frac{\partial P}{\partial Y} \cdot Y + P - W \cdot \frac{\partial L}{\partial Y} = 0 \\ \Leftrightarrow \quad & P \cdot \left(1 + \frac{\partial P/\partial Y}{P/Y}\right) = \frac{W}{\partial Y/\partial L} \\ \Leftrightarrow \quad & P \cdot \underbrace{\left(1 - \frac{1}{\eta}\right)}_{=\text{MR (marginal revenue)}} = \underbrace{\frac{W}{\lambda}}_{=\text{MC (marginal cost)}}, \end{aligned} \tag{3}$$

where $-\eta = (\partial Y/\partial P)/(Y/P) < 0$ denotes the price elasticity of output demand and λ denotes the marginal product of labor.

+ Note from (3) that the firm will choose a level of output such that $MR > 0$ and thus $\eta > 1$:

$$1 - \frac{1}{\eta} > 0 \quad \Rightarrow \quad 1 > \frac{1}{\eta} \quad \Rightarrow \quad \eta > 1.$$

+ From Equation (3) we have

$$P = \frac{MC}{1 - 1/\eta} \quad \Leftrightarrow \quad P = (1 + \theta) \cdot MC, \tag{4}$$

  with the mark-up parameter $\theta = 1/(\eta - 1)$.
+ The degree of pricing power of the firms, that is, the extent to which they can charge mark-ups on their goods and services, is related one-to-one to η: under perfect competition, we have that $\eta \to \infty$ and thus $\theta \to 0$. Also note that under $\eta > 1$ we have that $\theta \to \infty$ as $\eta \to 1$.
+ Moving from price levels to the rate of inflation, $\pi = \Delta P/P$, note from (3) and (4) that

$$\Delta P = \Delta\left[(1+\theta) \cdot \frac{W}{\lambda}\right] = \frac{\partial P}{\partial \theta} \cdot \Delta\theta + \frac{\partial P}{\partial W} \cdot \Delta W + \frac{\partial P}{\partial \lambda} \cdot \Delta\lambda$$

$$\Leftrightarrow \quad \Delta P = \frac{W}{\lambda} \cdot \Delta\theta + \frac{1+\theta}{\lambda} \cdot \Delta W + \left(-\frac{1}{\lambda^2}\right) \cdot (1+\theta) \cdot W \cdot \Delta\lambda$$

and thus: $$\pi = \frac{\Delta P}{P} = \frac{\frac{W}{\lambda}}{(1+\theta) \cdot \frac{W}{\lambda}} \cdot \Delta\theta + \frac{(1+\theta)/\lambda}{(1+\theta) \cdot \frac{W}{\lambda}} \cdot \Delta W - \frac{(1+\theta) \cdot \frac{W}{\lambda^2}}{(1+\theta) \cdot \frac{W}{\lambda}} \cdot \Delta\lambda$$

$$\Leftrightarrow \quad \pi = \underbrace{\frac{\Delta\theta}{1+\theta}}_{\substack{\text{inflation increasing}\\ \text{as growth of}\\ \text{mark-ups is rising}}} + \underbrace{\frac{\Delta W}{W}}_{\substack{\text{inflation increasing}\\ \text{as wage growth}\\ \text{is rising}}} - \underbrace{\frac{\Delta\lambda}{\lambda}}_{\substack{\text{inflation decreasing}\\ \text{as marginal product}\\ \text{of labor is rising}}} . \tag{5}$$

To obtain a complete picture about the rate of inflation, π, we turn to discussing the determinants of nominal wage growth, $\Delta W / W$.

Wage Bargaining

Nominal wages agreed upon in wage bargaining arguably should grow faster

+ the higher the expected rate of inflation,
+ the higher the growth rate of the marginal product of labor, and
+ the lower the level of unemployment, as worker turnover (that *ceteris paribus* always is costly for employers) then is more of a threat for employers, it under lower levels of unemployment being
  - easier for workers to find a new job, and
  - more difficult for employers to recruit.
+ We formalize these considerations about wage bargaining using the following simple linear specification:

$$\frac{\Delta W}{W} = \pi^{e} + \frac{\Delta \lambda}{\lambda} + \underbrace{\underbrace{\omega}_{\omega > 0} \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right)}_{\text{wage mark-up}} . \tag{6}$$

+ Note that the specification of the wage mark-up in Equation (6) reflects that unemployment from our Burns-Mitchell diagrams is countercyclical, and thus falls (rises) when actual output, Y, rises (falls) relative to long-run output, $\bar{Y}$.
+ Substituting from the wage growth equation (6) back into the firm-pricing implied equation (5) for the rate of inflation, we obtain:

$$\pi = \frac{\Delta \theta}{1 + \theta} + \pi^{e} + \underbrace{\omega}_{\omega > 0} \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) . \tag{7}$$

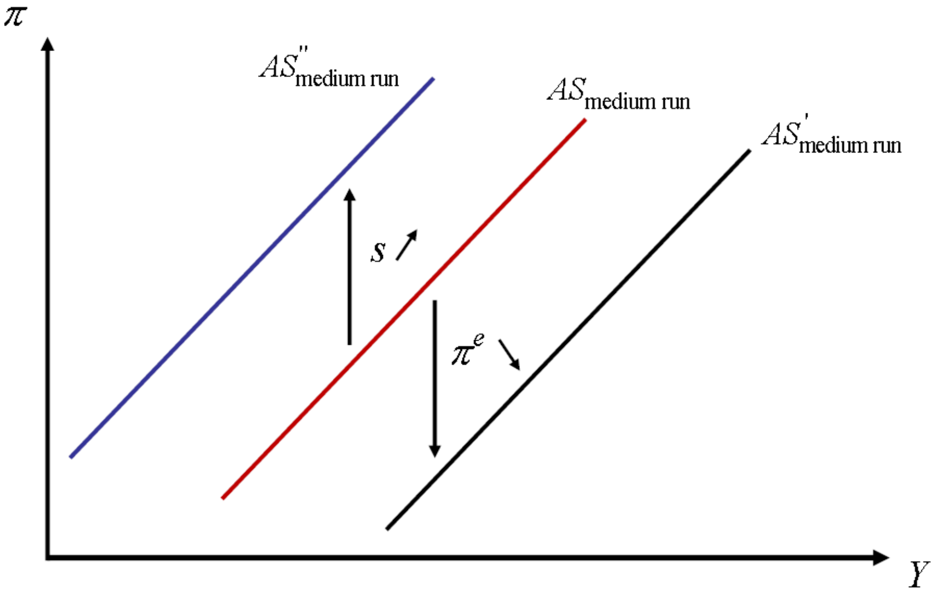
- We have not yet accounted for changes in non-labor costs of production, including intermediate inputs and physical capital. When the non-labor costs of production vary significantly over the business cycle, then it is often due to exogenous shocks: oil or other primary commodity price shocks, or foreign exchange market shocks raising the prices of imported components.
- We refer to such shocks as supply shocks, and append them to Equation (7):

$$\pi = \frac{\Delta\theta}{1+\theta} + \pi^e + \omega \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + s. \tag{8}$$

- The cyclical change of θ, the price mark-up, is ambiguous: In a boom, when actual output, Y, exceeds long-run output, $\bar{Y}$, then aggregate output demand is high, creating an opportunity to increase θ, but as competition among firms then tends to be higher as well, a reduction of θ could also be called for.
- For simplicity, our medium-run *AS*-curve in Equation (1) thus does not include a price mark-up:
- Note that the medium-run *AS*-curve shifts, for example, with changes in expected inflation, π^e, or with supply shocks, s.

$$\pi = \pi^e + \omega \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + s.$$

- Note that the medium-run *AS*-curve shifts, for example, with changes in expected inflation, π^e, or with supply shocks, s.



π
$AS''_{\text{medium run}}$
$AS_{\text{medium run}}$
$AS'_{\text{medium run}}$
$s \nearrow$
$\pi^e \searrow$
Y

Workbook for the Medium-Run AS-Curve

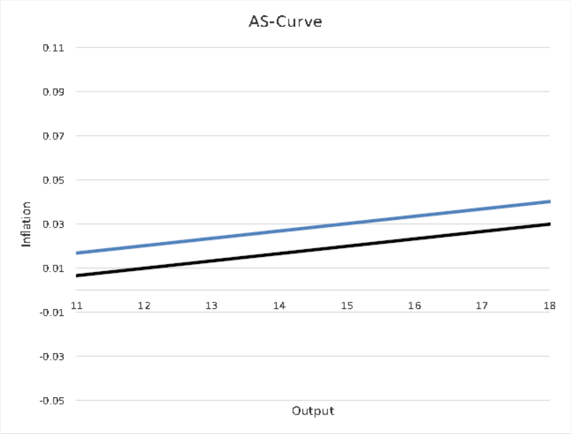
The workbook *AS-Curve.xlsm* allows to study all the determinants of the position and slope of the medium-run AS-curve.

AS-Curve: Specification

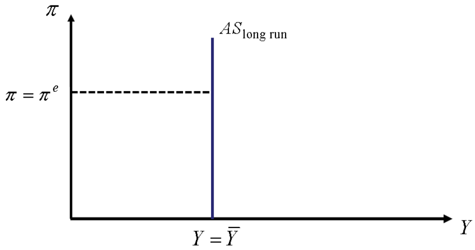
Parameters

	Initial	New
π^e	0.020	0.020
$\overline{Y}$	15.000	15.000
ω	0.050	0.050
s	0.000	0.010

AS-Curve: Graph



- Also note that Equation (1) implies that the *AS*-curve in the long-run must be vertical: In the long run, $Y = \bar{Y}$, and the economy has adapted to supply shocks (so that we might as well set $s = 0$ in the long run), and from Equation (1) we then have that $\pi = \pi^e$.



- An aside: Recalling once more that unemployment is countercyclical, we may also write Equation (1) as:

$$\pi = \pi^e - \omega_u \cdot \underbrace{\left(\frac{u - \bar{u}}{\bar{u}}\right)}_{\text{unemployment gap}} + s. \tag{9}$$

- Equation (9) is also known as the **Phillips curve**. This curve suggests that when actual unemployment, u, falls relative to the long-run level of unemployment, $\bar{u}$, then the rate of inflation rises (and vice versa). (Equation (9) does not suggest, though, that a higher rate of inflation, π, would lead to a lower rate of unemployment, u.)

To pin down medium-run macroeconomic outcomes, we will need to complement our analysis as to how the rate of inflation is determined as a function of output (as reflected in the medium-run *AS*-curve) by a relationship describing how output depends on the rate of inflation (wich will be called the ***AD-curve***).

We will obtain the *AD*-curve by substituting the real interest rate determined in the financial sector (summarized by the *TR*-curve, as applicable in the medium run) into the equation determining the level of output (summarized by the *IS*-curve).

The *AD*-curve will thus incorporate the short-run model's capturing of the **"real sector"** of the economy (the interaction between the aggregated demand for goods and services with the production of the goods and services), as well as its capturing of the **"financial sector"** of the economy (how interest rates are set through the interaction between savers, borrowers, the central bank and commercial banks).

Initially, our medium-run model (that we will call the *AS-AD* model), will abstract from "deep crisis" issues, specifically the zero lower-bound for the monetary policy rate and the endogenous component of the risk premium, and thus be a model of "normal business cycle" outcomes.

We will later consider "deep crisis" issues also, in an extended version of the *AS-AD* model, the extended *AS-AD* model.

Let us derive the *AD*-curve:

Medium-Run *TR*-Curve:
As in the medium run we allow for non-zero rates of inflation, the Taylor rule as formulated in Equation (60) of the chapter on "The Macroeconomy in the Short Run" is applicable:

$$R^{MP} = \underbrace{r_n^{MP}}_{=r_0^{MP}} + \pi^e + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right),$$

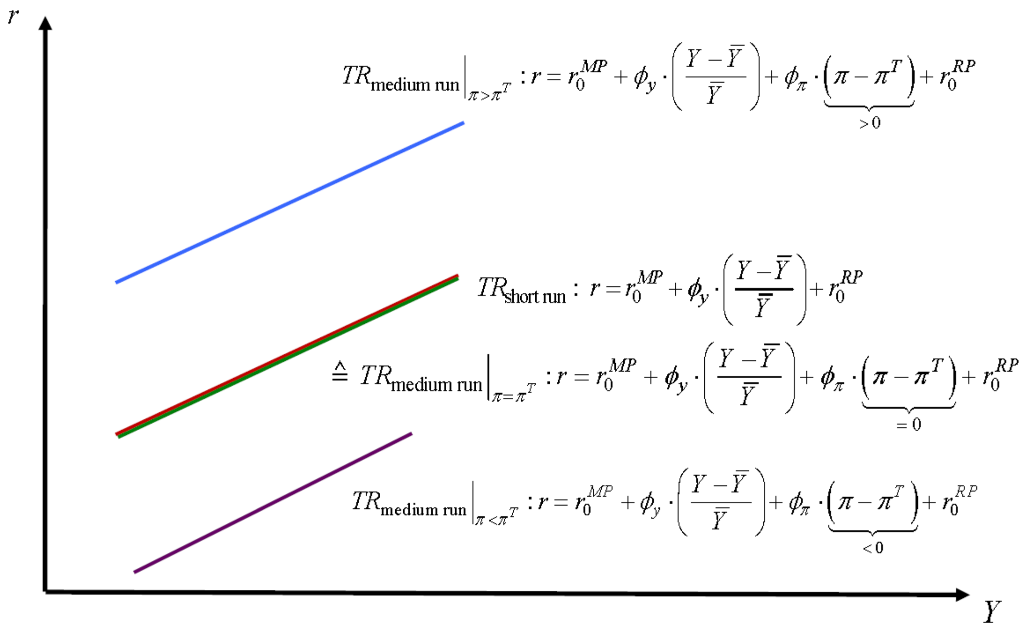
or, in terms of the real monetary policy rate,

$$r^{MP} \approx R^{MP} - \pi^e = r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right). \tag{10}$$

The medium-run *TR*-curve is thus given by:

$$r = r^{MP} + r_0^{RP} = r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi - \pi^T\right) + r_0^{RP}. \tag{11}$$

Let's Plotting the medium-run *TR*-curve:



r
$$TR_{\text{medium run}}\Big|_{\pi>\pi^T}: r = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \underbrace{\left(\pi-\pi^T\right)}_{>0} + r_0^{RP}$$
$$TR_{\text{short run}}: r = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + r_0^{RP}$$
$$\triangleq TR_{\text{medium run}}\Big|_{\pi=\pi^T}: r = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \underbrace{\left(\pi-\pi^T\right)}_{=0} + r_0^{RP}$$
$$TR_{\text{medium run}}\Big|_{\pi<\pi^T}: r = r_0^{MP} + \phi_y \cdot \left(\frac{Y-\bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \underbrace{\left(\pi-\pi^T\right)}_{<0} + r_0^{RP}$$
Y

We can use the same specification of the *IS*-curve as in the short-run model (as the short-run model's *IS*-curve involved only real quantities, and thus does not need to be adapted to the presence of non-zero rates of inflation). Equation (51) of the chapter on "The Macroeconomy in the Short Run" provided the *IS*-curve:

$$Y = \frac{A}{\underbrace{1-(C_y - TB_{im} - G_y)}_{=IS_0}} - \frac{C_r + I_r + TB_\varepsilon \cdot \varepsilon_r}{\underbrace{1-(C_y - TB_{im} - G_y)}_{=IS_1}} \cdot r \tag{12}$$

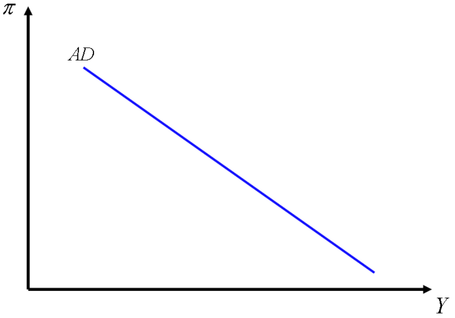
***AD*-Curve:** Combining the medium-run *TR*-curve (11) and the *IS*-curve (12), we obtain:

$$Y = IS_0 - IS_1 \cdot \left[r_0^{MP} + \phi_y \cdot \left(\frac{Y - \bar{Y}}{\bar{Y}} \right) + \phi_\pi \cdot \left(\pi - \pi^T \right) + r_0^{RP} \right]. \tag{13}$$

Equation (13) can be solved for the rate of inflation:

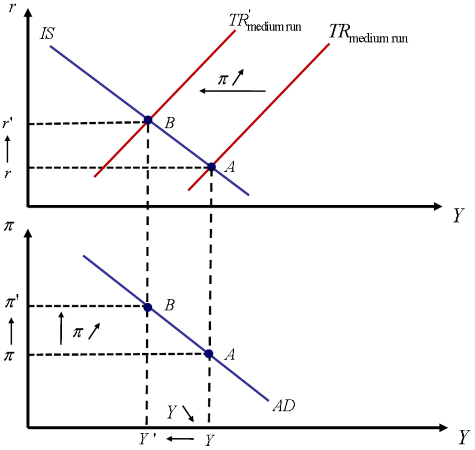
$$\pi = \frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP} \right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}} \cdot Y. \tag{14}$$

Equation (14) is the *AD*-curve, involving a negative relationship between output and inflation.

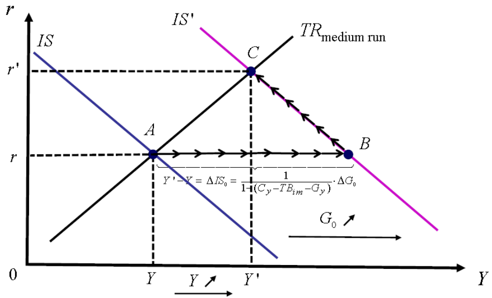


What is the economic rationale for the *AD*-curve being downward sloping?

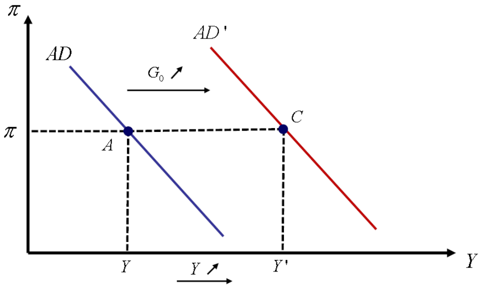
+ As the rate of inflation increases, the central bank, following the Taylor rule, raises the monetary policy rate.
+ The increase in the monetary policy rate causes (for a given risk premium) the real interest rate at which households and firms can borrow to increase.
+ Through the consumption, investment and exchange rate channels, this lowers output.



- Note that the *AD*-curve shifts when any of the parameters or exogenous variables entering the *IS*-curve and/or the medium-run *TR*-curve change.
- Consider, for example, an increase in G_0: This leads to a shift of the *IS*-curve.
- Graphing this in the medium-run *IS*-*TR* model diagram:



- Bringing this analysis to the *AD*-curve diagram:



- The move from Point *A* to Point *C* involves both the Keynesian multiplier effect (from Point *A* to Point *B* in the medium-run *IS*-*TR* diagram) and the crowding out effects due to the consumption, investment and exchange rate channels (from Point *B* to Point *C* in the medium-run *IS*-*TR* diagram).

Workbook for the AD-Curve

The workbook *AD-Curve.xlsm* allows to study all the determinants of the position and slope of the AD-curve.

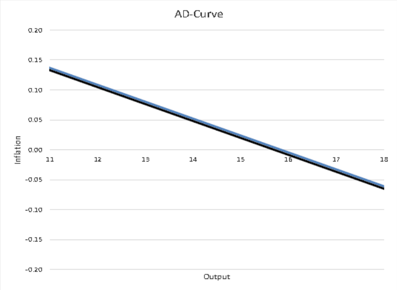
AD-Curve: Specification

Parameters

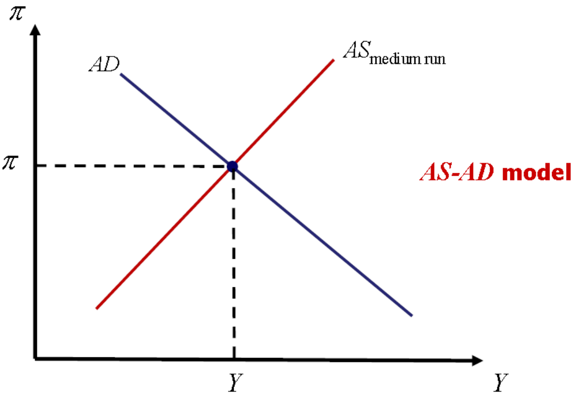
TR-Curve Parameters	Initial	New
r_0^{MP}	0.020	0.020
r_0^{RP}	0.010	0.010
ϕ_y	0.500	0.500
ϕ_π	2.000	2.000
$\bar{Y}$	15.000	15.000
π^T	0.020	0.020

IS-Curve Parameters	Initial	New
C_r	4.000	4.000
I_r	16.000	16.000
TB_ε	1.000	1.000
ε_r	6.000	6.000
c_y	0.600	0.600
TB_{im}	0.050	0.050
G_y	0.150	0.150
A	9.780	9.980
C_0	1.500	1.500
I_0	3.000	3.000
G_0	2.800	3.000
TB_0	0.000	0.000
ε_0	1.000	1.000
T	3.000	3.000
TB_{ex}	0.010	0.010
Y^*	100.000	100.000
T^*	30.000	30.000
r^*	0.030	0.030

AD-Curve: Graph



The medium-run macroeconomic outcome occurs at the intersection of the medium-run AS-curve with the AD-curve: it is the one combination of output and inflation that is compatible with the medium-run outcomes in both the real and the financial sector.



To appreciate better how the *AS-AD* model explains the functioning of the macroeconomy in the medium run, let us consider what happens following an increase in foreign income, Y^*. For such an analysis, we still need to pin down whether, and if so, how, inflation expectations respond to changes in the actual rate of inflation. This is as inflation expectations enter the medium-run *AS*-curve, which from Equation (1) is given by

$$\pi_t = \pi_t^e + \omega \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}} \right) + s_t.$$

Thus, if inflation expectations respond to changes in the actual rate of inflation, there will be a feedback loop from actual inflation to inflation expectations to actual inflation, and this needs to be taken into account. Two often-used specifications of how inflation expectations are formed are:

- **inflation expectations** are **static**:

$$\pi_t^e = \pi^e, \tag{15}$$

  inflation expectations then do not respond to changes in the actual rate of inflation.

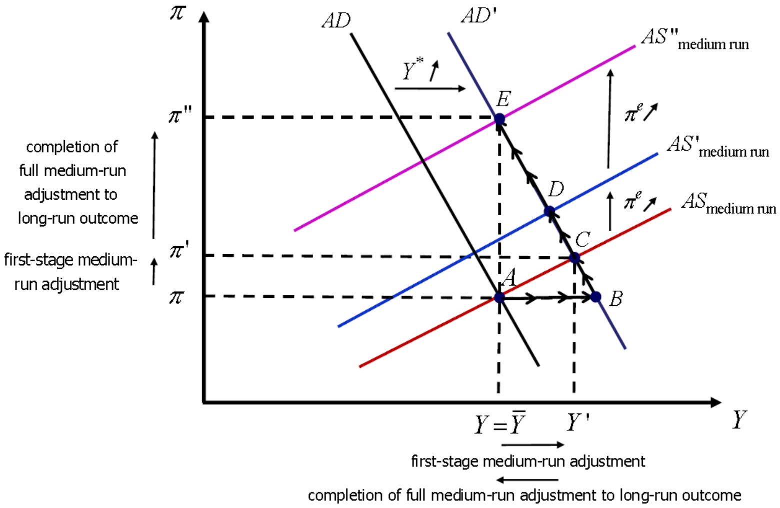
- **inflation expectations** are **adaptive** and adjust one-to-one to the previous period's actual rate of inflation:

$$\pi_t^e = \pi_{t-1}. \tag{16}$$

For now, let us choose the less restrictive specification, that of adaptive expectations. Substituting from (16) into the medium-run *AS*-curve, Equation (1), we obtain:

$$\pi_t = \pi_{t-1} + \omega \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}} \right) + s_t. \tag{17}$$

Set of adjustments following an increase in foreign income, Y^* (suppposing that initially output in the domestic economy, Y, is equal to long-run output, $\bar{Y}$):



- Starting from the original long-run outcome at Point *A*, after the increase in Y^* the *AD*-curve shifts to the right.
- The move from Point *A* to Point *B* reflects the short-run adjustments of the macroeconomy:
  - the initial increase in desired aggregate demand,
  - the resultant Keynesian multiplier effects, and
  - the crowding out of consumption, investment and the trade balance in response to the central bank in line with the Taylor rule raising the monetary policy rate.
- At Point *B*, we are off the medium-run *AS*-curve, however: As output is expanding, wage mark-ups rise, leading to accelerated growth of production costs and thus also accelerated growth of the prices of goods and services, that is, rising inflation.
- The rise in inflation causes the central bank, again in line with the Taylor rule, to further raise the monetary policy rate, causing further crowding out. For given inflation expectations, this crowding out occurs as we move along the *AD*-curve from Point *B* to Point *C*: Point *C* is a medium-run outcome in both the real and the financial sector.
- Under adaptive inflation expectations overall medium-run adjustment does not stop at Point *C*, though (it would stop there if inflation expectations were static).
- As actual inflation is increasing, under adaptive inflation expectations the inflation expectations start to rise as well, and the medium-run *AS*-curve in the next period of medium-run adjustment shifts upwards, implying higher rates of inflation for given levels of output.
- Starting from the original long-run outcome at Point *A*, after the increase in Y^* the *AD*-curve shifts to the right.

- The move from Point *A* to Point *B* reflects the short-run adjustments of the macroeconomy:
  - the initial increase in desired aggregate demand,
  - the resultant Keynesian multiplier effects, and
  - the crowding out of consumption, investment and the trade balance in response to the central bank in line with the Taylor rule raising the monetary policy rate.
- At Point *B*, we are off the medium-run *AS*-curve, however: As output is expanding, wage mark-ups rise, leading to accelerated growth of production costs and thus also accelerated growth of the prices of goods and services, that is, rising inflation.
- The central bank responds to this further rise of inflation, again raising the monetary policy rate in an attempt to curb the inflationary process. The implied further crowding out of consumption, investment and the trade balance is reflected in the move along the *AD*-curve from Point *C* to Point *D*. Point *D* is the medium-run outcome in both the real and the financial sector in the second period of adjustment.
- At Point *D*, relative to the starting point at *A*, we are still in an expansionary phase for output. As actual inflation has increased again and inflation expectations are adaptive, the inflation expectations continue to rise.
- The medium-run *AS*-curve thus continues to shift upwards, implying yet higher rates of inflation for given levels of output (leading the central bank, in its continued attempt to curb the inflationary process, to keep raising the monetary policy rate, in turn driving continued crowding out of consumption, investment and the trade balance).
- Inflation expectations only stop rising when the medium-run *AS*-curve has shifted so far upwards that it intersects the prevailing *AD*-curve at the initial, long-run level of output. At Point *E*, we have that $Y_t = \bar{Y}$, and thus from the medium-run *AS*-curve, Equation (17), it then holds that $\pi_t = \pi_{t-1} = \pi_t^e$ (as $s_t = 0$).
- At Point *E*, when medium-run adjustment has completed, the foreign income-driven aggregate demand stimulus has caused a higher rate of inflation; all short- and medium-run increases of the level of output in the domestic economy have been crowded out.

The medium-run macroeconomic outcome implied by the *AS-AD* model is obtained by combining the *AD*-curve, Equation (14), amended to include time subscripts,

$$\pi_t = \frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}} \cdot Y_t, \tag{18}$$

with the medium-run *AS*-curve, Equation (17),

$$\pi_t = \pi_{t-1} + \omega \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}}\right) + s_t.$$

These are two equations in two unknowns, the rate of inflation, π_t, and the output level, Y_t.
The level of output in the medium-run macroeconomic outcome is obtained from equating the *AD*- and medium-run *AS*-curve equations:

$$\underbrace{\frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}} \cdot Y_t}_{\text{from } AD\text{-curve (18)}} = \underbrace{\pi_{t-1} + \omega \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}}\right) + s_t}_{\text{from medium-run } AS\text{-curve (17)}}, \tag{19}$$

Note that (19) is one equation in one unknown, namely Y_t. Solving for Y_t:

$$\frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \pi_{t-1} + \omega - s_t = \left(\frac{\omega}{\bar{Y}} + \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}}\right) \cdot Y_t.$$

$$\Leftrightarrow \quad Y_t = \frac{\frac{\left(\phi_y + \phi_\pi \cdot \pi^T - r_0^{MP} - r_0^{RP}\right) \cdot IS_1 + IS_0}{\phi_\pi \cdot IS_1} - \pi_{t-1} + \omega - s_t}{\frac{\omega}{\bar{Y}} + \frac{\bar{Y} + \phi_y \cdot IS_1}{\phi_\pi \cdot IS_1 \cdot \bar{Y}}}. \tag{20}$$

Equation (20) gives us the *AS-AD* model-implied level of **output** in the medium-run macroeconomic outcome.
Having obtained the level of output, we can calculate the *AS-AD* model-implied rate of **inflation** in the medium-run macroeconomic outcome from the medium-run *AS*-curve, Equation (17),

$$\pi_t = \pi_{t-1} + \omega \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}}\right) + s_t,$$

$$\Leftrightarrow \quad \pi_t = \pi_{t-1} - \omega + s_t + \frac{\omega}{\bar{Y}} \cdot Y_t, \tag{21}$$

substituting for Y_t from Equation (20). Note that Equation (21) describes a dynamic process for the rate of inflation.

The real interest rate can now be calculated as well: Given the level of output and the rate of inflation, the medium-run *TR*-curve in Equation (11), amended to include time subscripts, yields the *AS-AD* model-implied **real interest rate** in the medium-run macroeconomic outcome:

$$r_t = r_0^{MP} + \phi_y \cdot \left(\frac{Y_t - \bar{Y}}{\bar{Y}}\right) + \phi_\pi \cdot \left(\pi_t - \pi^T\right) + r_0^{RP}, \tag{22}$$

with Y_t given by Equation (20), and π_t given by Equation (21).

Given output, inflation and the real interest rate, the medium-run macroeconomic outcomes of the other key macroeconomic aggregates, including consumption, investment, government expenditure, the trade balance and the exchange rate, may be retrieved from the *IS*-portion of the model.

Numerical Example:

While the algebra is tedious, a numerical example will be illuminating. Note from Equations (20) and (21) that we need to initialize the value of inflation to calculate a first medium-run macroeconomic outcome for, say, period $t = 1$. Suppose therefore that

we start out in period $t = 0$ in the long-run macroeconomic outcome, so that $Y_0 = \bar{Y}$, $r_0 = r_0^{MP} + r_0^{RP}$, and $\pi_0 = \pi^T$. Also suppose $A = 9.78$, $C_y = 0.6$, $G_y = 0.15$, $TB_{im} = 0.05$,

$TB_{ex} = 0.01$, $C_r = 4$, $I_r = 16$, $TB_\varepsilon = 1$, $\varepsilon_r = 6$, $\bar{Y} = 15$, $r_0^{MP} = 0.02$, $r_0^{RP} = 0.01$, $\phi_y = 0.5$, $\phi_\pi = 2$, $\pi^T = 0.02$, $\omega = 0.05$ and $s = 0$. Then for period $t = 1$ we obtain from Equation (20), under

$\pi_0 = \pi^T$, as the medium-run level of output $Y_1 = 15$. Using this value of Y_1 in Equation (21), we find that the medium-run rate of inflation in period $t = 1$ is given by $\pi_1 = 0.02$. Furthermore, from Equation (22), the medium-run real interest rate in period $t = 1$ for these values of Y_1 and π_1 is given by $r_1 = 0.03$. Now suppose that in period $t = 2$ the

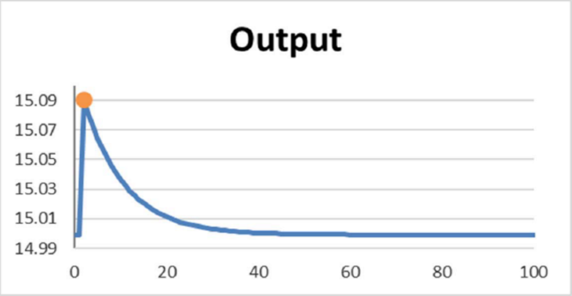
foreign income, Y^*, increases by 15, that is, $\Delta Y^* = 15$. As in the short-run analysis, this increase of foreign income implies a change in the autonomous component of aggregate demand by

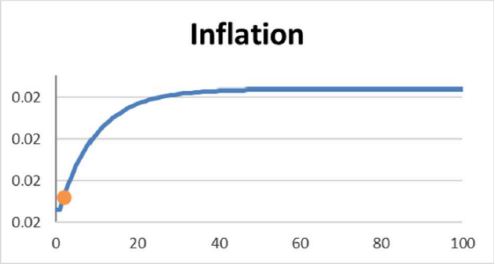
$$\Delta A = \frac{\partial A}{\partial Y^*} \cdot \Delta Y^* = TB_{ex} \cdot \Delta Y^* = 0.15.$$

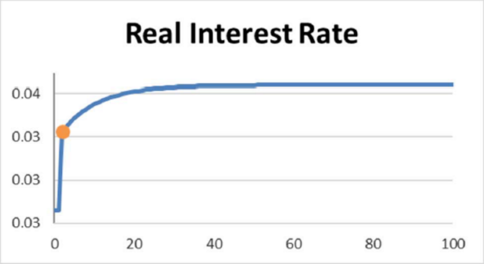
Using the implied new value of IS_0, $IS_0 = (A + \Delta A) \,/\, [1 - (C_y - TB_{im} - G_y)] = 16.55$, as well as $\pi_1 = 0.02$ in Equation (20), we obtain as the period $t = 2$ medium-run level of output $Y_2 \approx 15.094$. Using the (exact) level of Y_2 in Equation (21), we find the medium-run rate of inflation in period $t = 2$: $\pi_2 \approx 0.020$. Furthermore, from Equation (22), the medium-run real interest rate in period $t = 2$ for the (exact) values of Y_2 and π_2 can be calculated to be $r_2 \approx 0.034$ (following the Taylor rule, the central bank has increased the monetary policy rate, and so the real interest rate has risen).

Given that the (exact) value of the rate of inflation has increased from period $t = 1$ to period $t = 2$ and inflation expectations are adaptive, the adjustment process does not end in period $t = 2$. Using the same steps as for period $t = 2$, we may calculate the values of medium-run output, inflation and the real interest rate for all subsequent periods, $t = 3, 4, \ldots$, until medium-run adjustment is complete and we have reached the long-run levels.

Clearly, such calculations can be quite time-consuming, and we confine ourselves here to reporting the results, based on automated calculations within the workbook *AS-AD-Model.xlsm*. The following graphs show the full medium-run adjustment paths implied by Equations (20), (21) and (22):


Output
15.09
15.07
15.05
15.03
15.01
14.99
0
20
40
60
80
100


Inflation
0.02
0.02
0.02
0.02
0
20
40
60
80
100


Real Interest Rate
0.04
0.03
0.03
0.03
0
20
40
60
80
100

Workbook for the *AS-AD* Model

The workbook *AS-AD-Model.xlsm* allows to study how changes in the various parameters and exogenous variables of the *AS-AD*-model ...

AS-AD Model: Parameter Specification

	TR-Curve Parameters						IS-Curve Parameters																		AS-Curve Parameters	
Period	r_0^{NW}	r_0^{RP}	ϕ_y	ϕ_π	$\bar{Y}$	π'	c_r	i_r	TB_ε	s_r	c_y	TB_{im}	G_y	A	C_0	I_0	G_0	TB_0	s_0	T	TB_{ex}	Y^*	T^*	r^*	ω	s
0	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
1	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.780	1.500	3.000	2.800	0.000	1.000	3.000	0.010	100.000	30.000	0.030	0.050	0.000
2	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
3	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
4	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
5	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
6	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
7	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
8	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
9	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
10	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
11	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
12	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
13	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
14	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
15	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
16	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
17	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
18	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
19	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
20	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
21	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
22	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
23	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
24	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
25	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
26	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
27	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
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29	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
30	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
31	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
32	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
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34	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
35	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
36	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
37	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
38	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
39	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000
40	0.020	0.010	0.500	2.000	15.000	0.020	4.000	16.000	1.000	6.000	0.600	0.050	0.150	9.930	1.500	3.000	2.800	0.000	1.000	3.000	0.010	115.000	30.000	0.030	0.050	0.000

… affect output, inflation and the other endogenous variables in the medium-run macroeconomic outcome, …

AS-AD Model: Quantitative Results and AS-AD Diagram

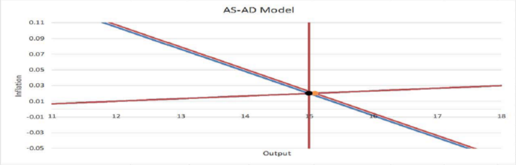
Medium-Run Macroeconomic Outcomes

	Period			
	0	1	2	1000
Inflation	0.020	0.020	0.020	0.023
Output	15.000	15.000	15.091	15.000
Real Interest Rate	0.030	0.030	0.034	0.036
Consumption	8.820	8.820	8.890	8.843
Investment	2.520	2.520	2.461	2.428
Government Expenditure	2.800	2.800	2.786	2.800
Trade Balance	1.100	1.100	1.223	1.215
Real Exchange Rate	1.000	1.000	0.978	0.965
Nominal Monetary Policy Rate	0.040	0.040	0.044	0.046
Nominal Interest Rate	0.050	0.050	0.054	0.056

	Current
Period	2

Plus

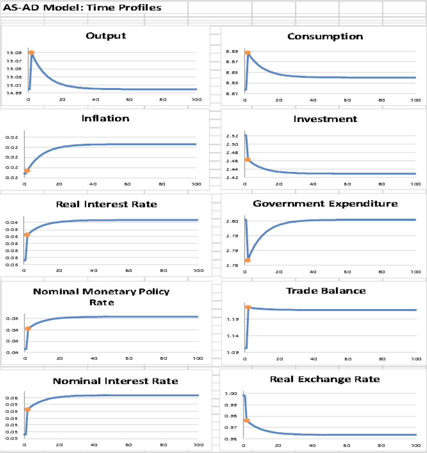
Minus



Rescale All Graphs

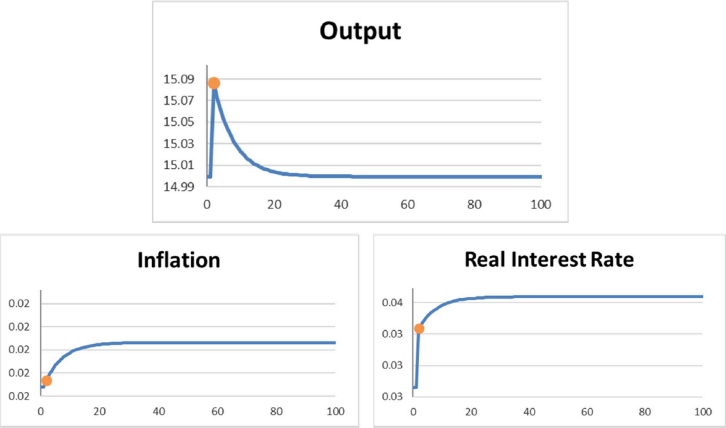
Horizons	
tmin	tmax
0	100

… and also graphs the transition dynamics (from the initialization onwards, and depicting the full medium-run adjustment):



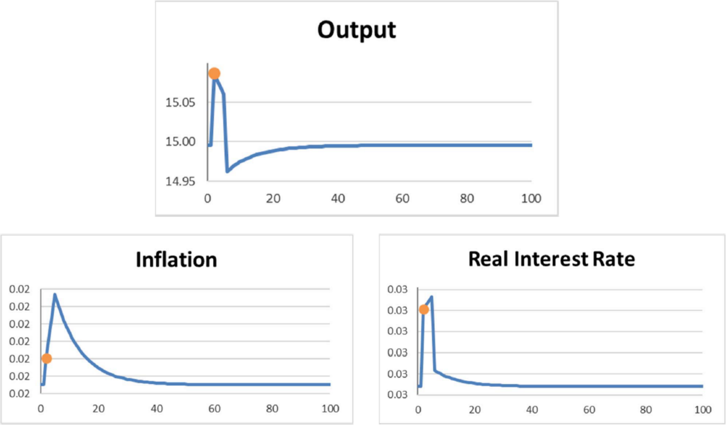
+ Under the scenario considered in the numerical example we have considered (a permanent positive stimulus to aggregate demand, following an initial medium-run macroeconomic outcome that is equal to the long-run macroeconomic outcome),
  - the level of output eventually returns to its long-run level, but
  - the rate of inflation and the real interest rate rise permanently.

  Thus, the cost to the economy of benefitting from a temporary expansion of output includes a permanent loss in the degree of price stability prevailing in the economy.
+ Everything else equal, if the central bank in its setting of the monetary policy rate was yet more responsive to deviations of actual inflation from target inflation, and we had, say, $\phi_\pi = 3$, then the time paths for output, inflation and the real interest rate would change to:

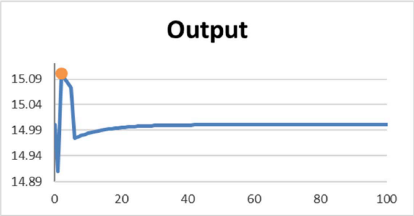

Output
15.09
15.07
15.05
15.03
15.01
14.99
0
20
40
60
80
100
Inflation
0.02
0.02
0.02
0.02
0.02
0
20
40
60
80
100
Real Interest Rate
0.04
0.03
0.03
0.03
0
20
40
60
80
100

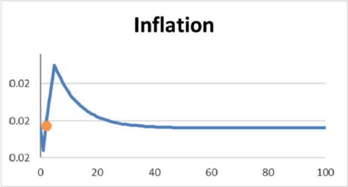
+ These graphs show the trade-off between
  - more sustained medium-run output gains facilitated by a ”less inflation responsive” central bank, coming at the cost of a stronger increase of long-run inflation,

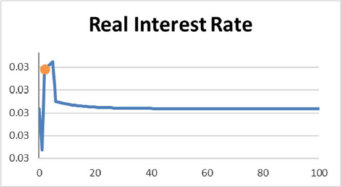
  and
  - the success of a ”more inflation responsive” central bank in reducing the rise of long-run inflation, coming at the cost of shorter-lived medium-run output gains.
+ All the scenarios considered so far involve the specification that the stimulus to aggregate demand persists forever (from $t = 2$ onwards for all periods). This is rather unlikely, though, taking into account also that we started from the long-run macroeconomic outcome.
+ Consider thus (setting $\phi_\pi = 2$ again) the case of a temporary aggregate demand stimulus, say an increase of foreign income, Y^*, by 15 that lasts for four periods only (from $t = 2$ to $t = 5$), before foreign income in period $t = 6$ is coming back down to its original level of $Y^* = 100$.
+ As the following graphs illustrate, it is then not just output that eventually converges back to its long-run level, but also the real interest rate; furthermore, inflation then converges back to the target rate of inflation, π^T.


Output
15.05
15.00
14.95
0
20
40
60
80
100
Inflation
0.02
0.02
0.02
0.02
0.02
0.02
0.02
0
20
40
60
80
100
Real Interest Rate
0.03
0.03
0.03
0.03
0.03
0.03
0
20
40
60
80
100

+ The adjustment path for output is particularly intriguing: following the peak output gain that occurs in period $t = 2$ (the first period of the temporary aggregate demand stimulus), the adjustment path involves a full business cycle (a contraction between $t = 2$ and $t = 6$, and then an expansion after $t = 6$ in the return to the long-run level of output).
+ The contraction occurs as in periods $t = 2, 3, 4$ and 5 we observe crowding out due to the central bank raising the monetary policy rate in its effort to curb rising inflation, and in period $t = 6$ due to aggregate demand falling back down to its original level (and this far outweighing the crowding-in effects caused by the central bank lowering the monetary policy rate in period $t = 6$ responding to the decline in aggregate demand).
+ The expansion occurs as from period $t = 7$ onwards aggregate demand is stable and the central bank keeps lowering the monetary policy rate while output is below its long-run level.
+ Using the workbook *AS-AD-Model.xlsm*, these dynamics can also be traced within the *AS-AD* model diagram.
+ Given what we have learned so far, it should not come as a surprise that
    - if the period $t = 1$ medium-run macroeconomic outcome was not equal to the long-run macroeconomic outcome,
    - then following a temporary aggregate demand stimulus,
    - output, inflation and the real interest rate would eventually converge back to their long-run/target levels (rather than their period $t = 1$ values).
+ To illustrate this point explicitly, consider a scenario in which in period $t = 1$ we have that $Y^* = 85$, so that the level of output in the period of the first medium-run macroeconomic outcome is below the long-run level of output. As in the previous scenario, foreign income then rises temporarily for four periods, from $t = 2$ to $t = 5$, to $Y^* = 115$, before it falls back down in period $t = 6$ to $Y^* = 100$.
+ The time paths of output, inflation and the real interest rate then are as follows, showing that these indeed converge back to their long-run/target levels:


Output
15.09
15.04
14.99
14.94
14.89
0
20
40
60
80
100


Inflation
0.02
0.02
0.02
0
20
40
60
80
100


Real Interest Rate
0.03
0.03
0.03
0.03
0.03
0
20
40
60
80
100